Chapter 2 — What Makes Capital Capital?

The Analytical Foundations of Capital Theory

2.1 Introduction

The objects economists classify as capital have changed considerably; the logic by which they are classified has not. Physical assets, human capabilities, knowledge, organisational resources, environmental assets and social relationships have each been admitted to the category, and in every case the argument turned on the recognition that the resource shared a set of economic characteristics with what was already inside.

This chapter identifies those characteristics. It does not review the individual categories, since the point is not what has been admitted but on what grounds, and the grounds must be established independently before the question of Chapter 3 can be asked without begging it. Eight properties are set out below. They are stated as a general test, applicable to any candidate resource, and Chapter 4 applies them to institutional capabilities without modification.

2.2 Stock and Investment

The first two properties concern what capital is and where it comes from.

Capital is a stock, which is to say that it exists at a point in time rather than being generated over a period. Income, output, investment and services are flows. The distinction is elementary and it is routinely lost, with consequences for both economic analysis and public policy, because the two are easily conflated in ordinary language: we speak of a country's education as though it were both the system and the schooling it delivers. Capital is the accumulated productive resource and not the services it yields. A machine is not the products it manufactures; human capital is not labour income. Any new category must therefore first be shown to be an accumulated stock capable of producing future services rather than the services themselves.

Capital also does not arise spontaneously. It results from investment, expenditure on infrastructure, machinery or equipment in the physical case; education, training and health in the human case; conservation or restoration in the natural case; management systems and organisational capability in the organisational case. Investment precedes formation in every category.

But investment does not guarantee formation, and the gap between them matters more than it is usually allowed to. Investments fail. Some yield only temporary benefits. Others produce assets that are obsolete before they are complete. Economic theory therefore distinguishes expenditure from successful capital formation, and the distinction bears with particular force on public institutions, where large outlays do not reliably produce durable capability. Chapter 5 returns to this at length.

> **Property 1.** Capital exists as an accumulated stock capable of generating future flows of productive services.
>
> **Property 2.** Capital is created through successful investment that results in durable, productive assets.

2.3 Durability and Productive Services

The next two properties concern how long capital lasts and what it does while it lasts.

Durability is what distinguishes capital from consumption. A training programme completed yesterday may raise productivity for decades; a bridge may remain in service for fifty years; a statistical information system may continue generating usable information long after its implementation. This does not imply permanence, every form of capital eventually deteriorates, but the economic contribution extends beyond the period in which the investment was made. That persistence is what justifies treating the expenditure as the acquisition of an asset rather than as a current cost.

What the asset does over that life is generate services, which is arguably its defining characteristic. The value of capital lies not in its existence, but in the flow it continuously provides: machinery produces manufacturing services, education produces skilled labour services, ecosystems produce environmental services, organisational systems produce coordination services. Two organisations holding identical capital stocks may perform quite differently if their capital yields different quantities or qualities of service, which is why performance depends on the flow rather than on the stock alone, and why measuring one as a proxy for the other, a temptation Chapter 9 addresses directly, is hazardous.

> **Property 3.** Capital possesses a durable productive life extending beyond the initial investment period.

> **Property 4.** Capital generates recurring flows of productive services.

2.4 Maintenance, Depreciation, Complementarity and Productivity

The remaining four properties concern what happens to capital over time and what determines whether it produces.

No form of capital remains productive indefinitely without **maintenance**. Infrastructure requires repair, human capital continued learning, natural capital conservation, organisational capital adaptation. Maintenance is therefore a component of capital preservation rather than an operational detail subordinate to investment, since failure to maintain reduces productive capacity as surely as failure to invest prevents its creation. Accumulation and maintenance are inseparable.

Capital also **depreciates** independently of neglect. Physical assets wear out, knowledge becomes obsolete, skills decline, organisations lose effectiveness, environmental assets degrade. The mechanisms differ across categories (physical deterioration, technological obsolescence, institutional decay, inadequate maintenance) but the principle is universal, and any account ignoring it will overestimate the capacity actually available.

Capital is **complementary**. Physical capital requires skilled workers; human capital depends on organisational systems; digital infrastructure requires legal frameworks. The productivity of one form frequently depends on the presence of others, which is why identical investments generate dissimilar outcomes across countries: effectiveness is a function not only of quantity but of the system within which the asset sits.

And capital exists, ultimately, because it raises **productive capacity**, because it increases the ability of an economy or organisation to convert inputs into outputs more efficiently. This is the economic justification for treating any resource as capital at all. A resource satisfying the preceding seven properties but raising no productive capacity would be a curiosity rather than an asset.

> **Property 5.** Capital requires continuous maintenance to preserve its productive capacity.

> **Property 6.** Capital is subject to depreciation through time.

> **Property 7.** Capital generates its highest returns through interaction with complementary forms of capital.
>
> **Property 8.** Capital enhances productive capacity by increasing the efficiency or effectiveness of production processes.

This eighth property needs a qualification, without which the test would chase its own tail. It admits production among the criteria of admission, whereas Chapters 7 and 9 treat production as the quantity the stock is required to *predict*: a test demanding that services be observed before the asset is admitted could no longer serve to explain those services. The property must therefore be read strictly *ex ante*, and it concerns design characteristics observable before any load is placed on the asset: is it designed to reduce the unit cost of an administrative output, to raise the quality of a decision, or to make possible an operation that was not? An identity system satisfies the property by its design, before it has registered a single person. Whether it actually produces belongs to Level III of Section 2.5 and has no place in qualification.

2.5 The General Test, and Its Three Levels

The forms of capital recognised in economics differ enormously in their physical nature and in the literatures that study them, yet they share a consistent analytical structure. A resource may be analysed as capital when it exists as an accumulated stock; results from investment; possesses durability; generates recurring productive services; requires maintenance; depreciates over time; interacts with complementary forms of capital; and enhances productive capacity.

These eight conditions define no particular category. They define what separates capital from other economic resources, and they are neutral as to what might satisfy them.

They should not, however, be read as a flat checklist, because they do not all do the same kind of work. Some establish whether an object is capital at all; others describe how it behaves once it is; others still govern whether it produces anything. Organising them hierarchically avoids a confusion that recurs throughout this book, between an asset that does not exist and an asset that exists without producing.

Level I — Qualification. Whether an object is capital turns on four conditions: that it constitutes an identifiable stock, that it is durable beyond the immediate period, that it possesses productive capability, and that it can generate future flows of services. These are the admission criteria, and failing any one of them is disqualifying.

Level II: Dynamic behaviour. Once an asset qualifies, it may be formed, accumulated, maintained, upgraded, appreciated, depreciated, rendered obsolete, impaired, stranded, retired or destroyed. These are life cycle properties rather than further tests. An asset that has depreciated badly is still capital; it is capital in poor condition, which is a different finding with a different remedy.

Level III — Productive deployment. Whether a qualifying stock actually produces services depends on several things: the condition it is in, the share of it actually drawn upon, and also the presence of complements, governance and institutional context. The last three are set aside here, Section 7.5 returns to them with the full production function. In this reduced form, the services delivered in a period are the product of what the state holds, the condition it is in, and how much of it is used:

$$S_t = K_t \, q_t \, u_t$$

Which reads: services delivered = what you hold × the condition it is in × the share of it you use.

S_t is the flow of services produced in period t. K_t is the qualifying stock held at that moment. q_t is its productive quality (condition, currency, degree of integration) and u_t the share actually drawn upon. The stock is a level and services a flow; q_t and u_t are proportions bounded between 0 and 1. The product $K_t q_t$ is potential service capacity, and u_t the fraction of it realised.

The three terms multiply rather than add. None of them compensates for weakness in another: a stock in good condition used at a quarter of its capacity returns barely a fifth of what it could, however large it is. Three distinct failures follow, calling for opposite responses. A state with small K must build. A state with low q built and then let the stock decay: it must maintain and upgrade. A state with low u has built, holds a stock in working order, and is not using it, further investment there would be spent against the wrong constraint.

This is what the three-level structure exists to preserve. An asset may satisfy every Level I criterion, behave at Level II exactly as capital behaves, and still produce almost nothing. An asset that does not exist and an asset that exists without producing are different findings.

That neutrality is what makes the next step legitimate. Governments invest continuously in institutional capabilities, and many of those capabilities appear, on inspection, to accumulate through investment, persist through time, require maintenance, deteriorate when neglected, generate recurring services and raise the productivity of other public resources. They are nonetheless rarely analysed in the language of capital.

This observation establishes nothing yet. It identifies a possibility, and possibilities of this kind have a poor record in economics when asserted rather than tested, the looser extensions of capital theory have been criticised precisely for asserting a family resemblance and declining to demonstrate one. Chapter 3 defines the category the test will be applied to, and Chapter 4 applies it, including to candidates that fail.

2.6 The Objection This Test Answers

Section 1.2 noted that several extensions of the idea of capital have been criticised for the looseness of the analogy on which they rest, and that social capital has been criticised most sharply of all, by Arrow [5] and by Solow [28]. The objection is worth stating precisely, because this chapter exists in order to answer it.

It has three parts. Calling something capital does not make it capital: the thing must behave as capital behaves. And a quantity behaves as capital only if it arises from a deliberate act of investment carrying an identifiable opportunity cost; if the stock can be distinguished from the returns it produces; and if it depreciates in a way that can be observed. Failing that, the word supplies not an analysis but a metaphor, and the metaphor then licenses inferences that nothing supports.

This book holds the objection to be sound. That is why it does not open by asserting that institutions constitute capital, but with a test that most candidates fail.

Three features distinguish Institutional Capital from the cases Arrow and Solow set aside. It is created by identifiable investment, attributable to a budget and carrying a measurable opportunity cost: a digital identity system is built, it does not accrete. Its stock is separable from its returns and observable independently of them, this is the work of Chapter 9, and it is what makes the theory refutable rather than circular. And it depreciates at a rate that depends on maintenance, which is an expenditure traceable in public accounts; depreciation is therefore not postulated but predicted, and Chapter 6 states through which mechanisms.

One concession must nonetheless be made, and it matters more than the answer. The objection cannot be disposed of in general, but only instance by instance. This is why qualification applies to the asset and not to the category: two tax administrations with identical statutes can receive different verdicts, and Chapter 4 examines three candidates that fail the test. A theory admitting every institution to the rank of capital would deserve precisely the criticism levelled at social capital. The eight-property test is not an entry formality: it is the device by which this theory accepts being limited.

Part II — Conceptual Framework

Chapter 4 — Why Institutional Capital Qualifies as Capital

Demonstrating the Economic Properties of Institutional Capital

4.1 Introduction and Strategy

Chapter 3 introduced Institutional Capital as the accumulated stock of durable institutional assets generating institutional services. Introducing a category, however, is the easy part. The demanding task is to establish whether it belongs within the family of capital recognised in economic theory, and that is what this chapter attempts to do.

The strategy is to propose no new criteria. The eight properties of Chapter 2 were derived independently of institutions, precisely so that applying them here would not beg the question, and the demonstration proceeds by asking of each the same thing: do certain institutional capabilities exhibit this characteristic? Where the answer is affirmative across all eight, the category qualifies.

Two features of what follows are worth noting in advance. The properties are taken in pairs rather than individually, since they group naturally, and the argument is clearer that way. And the chapter does not end with the affirmative verdict, Section 4.4 runs three further candidates through the same test and rejects two of them, because a test that admits everything establishes nothing about what it admits.

4.2 Applying the Eight Properties

Stock and investment. A national statistical office is not recreated every year; a digital identity system does not disappear at the close of a budget cycle. A tax administration accumulates databases, procedures, information systems, organisational routines, trained personnel, and operational knowledge over many years, existing at any given moment as a productive resource whose services may fluctuate while the underlying capability persists. Nor do such capabilities arise spontaneously: governments invest in legislation, organisational design, infrastructure, information systems, training, platforms and administrative modernisation, and where those investments succeed, they create durable assets. The word "succeed" carries weight that Chapter 5 makes explicit.

Durability and services. Civil registration systems, judicial institutions, central banks, audit institutions, statistical agencies, and identity infrastructure continue to produce services over decades. What they produce is a recurring flow (identity verification, tax administration, procurement management, financial reporting, statistical production, regulatory enforcement, interoperability, policy implementation) without which governments cannot function.

Maintenance and depreciation. Information systems require upgrades, database maintenance, revision of legal frameworks, retraining of civil servants, and adaptation of procedures to evolving circumstances. Deterioration also proceeds independently of neglect, through technological obsolescence, organisational decay, corruption, political instability, fragmentation and the loss of institutional memory, reducing productive capacity even where infrastructure remains intact and the organisation remains formally in existence. Chapter 6 develops both at length, since the invisibility of institutional depreciation is among this book's principal subjects.

Complementarity and productive capacity. Digital identity requires digital infrastructure; revenue administration depends on human capital; procurement depends on legal institutions. Institutional capital interacts continuously with physical, human, financial, technological, and organisational capital, and its productivity depends in part on these complementarities, which is why identical investments produce dissimilar results across countries. And it exists to increase productive capability: to enable governments to collect taxes more effectively, deliver services more efficiently, coordinate, regulate, implement, respond to crises and support economic activity. Rather than producing market goods, it increases the productive capacity of the public sector itself.

Proposition 4.1. Institutional Capital exists as an accumulated stock of productive institutional capabilities.

Proposition 4.2. Institutional Capital is created through sustained institutional investment.

Proposition 4.3. Institutional Capital has a productive life that extends beyond the period of initial investment.

Proposition 4.4. Institutional Capital generates recurring institutional services.

Proposition 4.5. Institutional Capital requires continuous maintenance to preserve its productive capacity.

Proposition 4.6. Institutional Capital is subject to depreciation through institutional, organisational, and technological change.

Proposition 4.7. Institutional Capital generates its highest returns when combined with complementary forms of capital.

Proposition 4.8. Institutional Capital increases the productive capacity of public institutions.

4.3 Synthesis

Institutional Capital satisfies each of the eight properties identified in Chapter 2.

General Property of Capital	Institutional Capital
Accumulated stock	✔
Created through investment	✔
Durable	✔
Generates productive services	✔
Requires maintenance	✔
Subject to depreciation	✔
Complementary	✔
Enhances productive capacity	✔

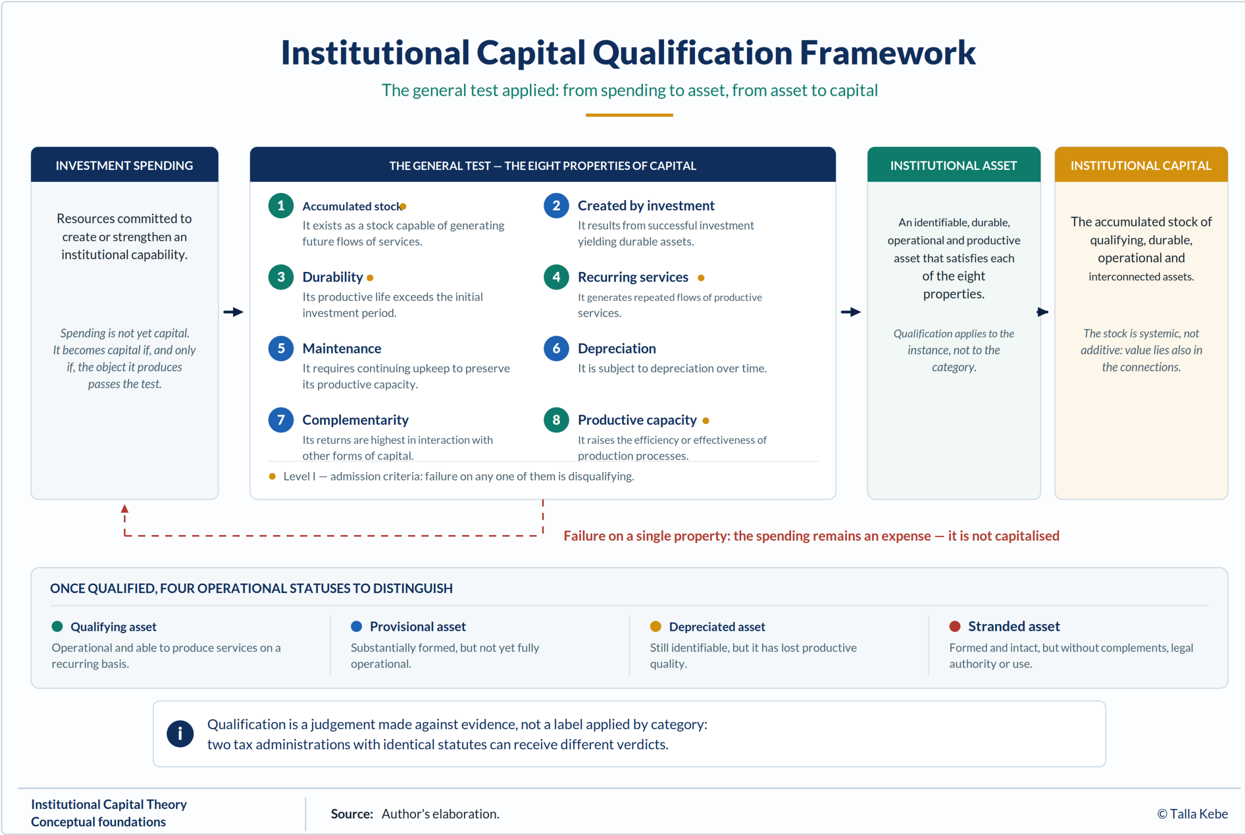

Figure 3: **The Institutional Capital Qualification Framework.** The general test applied: expenditure failing any of the eight properties remains an expense rather than being capitalised, together with the four operational statuses of an asset that passes. Source: author's conceptual framework.

A table of this kind is easy to produce and easy to distrust, and the reader is right to ask what it would take for one of the marks to be absent. Two things should therefore be said about what the verdict does and does not establish.

It does not establish that every institution constitutes capital. It establishes that a specific subset of institutional capabilities exhibits the characteristics traditionally associated with productive capital, a subset whose boundaries the next section tests directly.

A last clarification concerns a property frequently associated with Institutional Capital that does not appear among the eight, and which is better placed explicitly than left ambiguous: non-rivalry. Non-rivalry is not a condition of admission to capital, physical capital is rival and qualifies without difficulty, but it is a characteristic feature of Institutional Capital: a civil registry, an identity foundation or an interoperability standard can be drawn on simultaneously by many administrations without one body's use reducing its availability to another. It is this property that makes the returns to foundational assets diffuse and hard to attribute to any single service, a point to which Section 8.4 returns.

Nor does it establish that any particular institution in any particular country qualifies. The properties are satisfied by the class; whether a given revenue authority or registry satisfies them is a question about that authority or registry, answerable only on the basis of evidence. Chapter 5 explains why two administrations with identical statutes may receive different verdicts.

4.4 What Does Not Qualify

A test that admits everything tests nothing. The preceding sections returned eight affirmative verdicts, and a reader entitled to be sceptical will ask whether the framework can return any other answer. It can, and three candidates show how.

A constitutional principle does not qualify. The separation of powers is durable, consequential, and important, but it fails in the second and fourth properties. It is not accumulated through investment in any meaningful sense, a constituent act established it, and no quantity of subsequent expenditure increases the stock a country holds of it. And it generates no institutional services itself: it constrains and licenses the production of services by other assets, which is what a rule does. The verdict is not that it is unimportant, but that the apparatus of capital adds nothing to its analysis and would obscure what is distinctive about it.

A national development plan does not qualify, and the reason is instructive. A five-year plan is produced through substantial investment, is maintained in written form, and coordinates action across ministries. It fails at durability and depreciation. Its productive life does not extend beyond its horizon because the plan is consumed in use: its content is a current allocation of priorities, and priorities that have been superseded have no residual productive capacity. Nor does it depreciate in the sense that the framework requires; it expires, and no amount of maintenance can preserve it. What may qualify is the *planning system*: the analytical staff, costing models, data pipelines, and coordination routines that repeatedly produce plans. The plan is the output; the planning system is the asset. Distinguishing them is exactly the work the framework is for.

A newly established anti-corruption commission has yet to reach a verdict. It has been created through deliberate investment, has statutory permanence, and is intended to generate recurring services, satisfying the second and fourth properties on its face. Whether it satisfies durability cannot be determined at establishment, for the reason given in Chapter 5: investment produces an asset only where it produces a capability that is durable, operational, and repeatedly used. A commission that is defunded, captured or reduced to formal existence will have consumed investment without creating capital. Isomorphic mimicry is the standing risk, since anti-corruption commissions are among the most widely copied institutional forms in the world and among the most frequently copied without function.

Two consequences follow for the rest of the book. Qualification is a judgement on the basis of evidence rather than a label applied by category, so two revenue authorities with identical statutes may reach different verdicts. And the population of Institutional Capital in any country is smaller and harder to enumerate than the illustrative lists in Chapter 3 imply. Those lists name asset types that typically qualify, and whether a particular instance qualifies is an empirical question.

4.5 Qualification, Formation, and Operational Status

The three verdicts in the preceding section point to a distinction that the framework needs but has so far left implicit. Qualification is not the same question as formation, and neither is the same as whether an asset is currently working.

An asset qualifies when it is identifiable, durable, operational, productive and capable of generating recurring institutional services. Whether it was well or badly formed is a separate matter, taken up in Chapter 5. Whether it is presently producing anything is a third matter again, and four operational categories are worth distinguishing:

- A **qualifying asset** is operational and capable of recurring service production.

- A **provisional institutional asset** is substantially formed but not yet fully operational, the anti-corruption commission of Section 4.4, and a great many systems in their first years.
- A **depreciated or impaired asset** remains identifiable but has reduced productive quality. This is the condition Chapter 6 is about.
- A **stranded institutional asset** has been formed but cannot generate services, because critical complements, legal authority or utilisation are absent. A registry built without the legal basis to share its data is stranded; so is a platform nobody has been required to use.

The last category is worth dwelling on, since it is the one governments most often fail to recognise. Stranding is not a failure of investment, the asset exists and was built as specified. It is a failure of the surrounding system, and further investment in the asset itself will not release the value locked inside it.

One further distinction follows. Operational readiness describes whether an asset *can* produce services; utilisation describes the extent to which it is actually deployed. Services realised are therefore a product of both, applied to the stock:

$$IS_t = IC_t\, q_t\, u_t$$

Which reads: institutional services delivered = capital held × the condition it is in × the share of it used.

IS_t is the flow of institutional services produced in period t, IC_t the stock of Institutional Capital, q_t its productive quality and u_t its utilisation, the last two being proportions bounded between 0 and 1.

This is the institutional case of the general form given in Section 2.5 — operational readiness carried by q_t and deployment by u_t — and it has an immediate measurement consequence taken up in Chapter 9: a state observing low service output cannot tell, from that observation alone, whether it holds too little capital, capital in poor condition, or capital it is not using.

4.6 A Documented Stranded Asset

The category the preceding section has just defined is the one governments acknowledge least readily, and a framework that proposes it should produce at least one instance. The example taken here has a property few candidates possess: the verdict is delivered not by the author but by the audit institution of the state concerned, on specific dates and in published terms.

The South African government's Integrated Financial Management System, the IFMS, follows from a Cabinet decision of 2005. It was to replace fragmented and ageing financial, human resource and procurement systems across national and provincial government. Two decades later it is not in service.

The facts are those reported in the hearings of Parliament's public accounts committee [61]. Some 1.7 billion rand had been spent against a projected envelope of 4.2 billion. The Auditor-General classified the associated spending as fruitless and wasteful for five consecutive financial years from 2017–18, a classification the National Treasury formally disputed, holding it to be the discharge of a contractual commitment. A first version of the programme was abandoned in favour of a second, the contractor claiming 800 million rand in damages, negotiated down to 383 million. Direction was shared between three entities, the Treasury, the public service department and the state information technology agency. And the Treasury was paying some 68 million rand a year in maintenance on a system that had not been brought into service.

Why this asset is stranded rather than provisional or depreciated. A provisional asset is substantially formed but not yet fully operational; that description assumes a horizon, and twenty years exhausts it. A depreciated asset was productive and lost quality; this one never produced the service for which it was intended. What was built exists, has been paid for, is maintained, and bears no load.

What this case establishes that the theoretical argument alone could not. Three things.

The first concerns maintenance. Chapter 6 will argue that upkeep lowers the depreciation rate of a stock. Here, a maintenance expenditure is incurred year after year on an asset not established to constitute a stock producing services. The expenditure is real and probably contractually owed; it nonetheless lowers the depreciation rate of no capital in service. It is a carrying cost, and distinguishing it from maintenance proper follows directly from the distinction between formation and qualification.

The second concerns the conditions of formation. Direction split across three entities with no single authority corresponds exactly to the organisational absorption and continuity determinants that Section 5.3 identifies as components of formation efficiency. The investment was made; the conversion did not occur. This is Proposition 5.1 observed in a case where none of the usual explanations is available: no crisis, no budgetary squeeze, no state fragility. The country has an established treasury, an independent audit institution that publishes, and an active parliamentary oversight committee that took up the matter five times.

The third is the most important, and it lies in the accounting disagreement itself. The Auditor-General holds the spending to be without value received; the Treasury holds it to be the discharge of a contractual commitment. Both positions are defensible within their own frame, and the disagreement persists for a reason that must be stated exactly. It is not that public accounting is unaware of assets of this kind: the software layer of the system falls under IPSAS 31 [41] and capitalises without difficulty. It is that no category covers an asset regularly procured, regularly maintained, and *never brought into load*, that is, an asset whose service capacity, the condition of recognition, was never established. Neither “asset” nor “expense” nor “waste” describes the situation correctly, and it is this edge of the existing category, not its absence, that the dispute exposes. It is that gap the category of stranded institutional asset is meant to fill, and a five-year dispute between two competent institutions attests to it better than an argument can.

What this case does not establish. It does not settle the dispute, and this book has no business doing so: the qualification proposed here is analytical, not legal or accounting. Nor does it support any conclusion about how frequently stranding occurs, which a single case cannot measure. Section 9.3 indicates which indicators would identify it systematically, an asset with high installed coverage and no utilised coverage, and the fact that no statistical system collects them today is, consistent with Section 1.4, what should have been expected.

4.7 Relation to Existing Accounts of State Capacity as Investment

The demonstration above would be considerably less interesting if economics had never treated state capabilities as accumulated investment. It has, and the closest existing account should be stated plainly before the qualification principle is asserted.

Besley and Persson [8], [9] analyse fiscal capacity, the ability to raise broad-based taxes at reasonable cost, and legal capacity, the ability to support markets and enforce contracts, not as fixed endowments but as stocks that governments choose to invest in. In their framework, the investment path is determined by expected future returns and by the cohesiveness of political institutions, and the two capacities are complementary: investment in legal capacity that expands formal-sector employment raises the return to investment in fiscal capacity, and the relationship runs in both directions. They also construct a cross-country index of these capacities and relate it to income and to political violence.

Several elements of the present framework are therefore not novel and should not be presented as such. That public capabilities are created through investment, that they accumulate, and that they are complementary are propositions with an established home in political economy. Propositions 4.2 and 4.7 above are consistent with that work rather than departures from it.

Four features distinguish Institutional Capital Theory from this account.

Breadth of the asset class. Besley and Persson [8] model two capacities. The framework developed here extends to statistical systems, civil registration, procurement platforms, case management, digital identity, and interoperability, the registry and platform layer on which contemporary administration depends, and which their two-capacity structure does not represent. Whether this breadth is a gain or a loss of analytical discipline is a fair question; Core Proposition 2 exists to answer it by specifying an admission test rather than an open list.

Four features distinguish institutional assets from tangible ones, and they overlap with those Haskel and Westlake [16] systematised for intangibles: non-rivalry, spillovers, poor collateral value, and sensitivity to financing conditions. The first three explain why returns are diffuse and investment structurally insufficient; the fourth is the subject of Chapter 13. That these features are shared with intangibles generally strengthens rather than weakens the argument, since it places Institutional Capital within a class whose economics is already documented.

Depreciation and the life cycle. Their framework is an investment model. It has no counterpart to institutional depreciation, the maintenance gap, or erosion, the state in which depreciation persistently exceeds renewal. Chapter 6 develops these, and they are the sharpest single difference between the two accounts. A state can invest continuously and still lose capacity; an investment model alone cannot represent that outcome.

The wedge between investment and formation. Their model assumes that investment produces capacity. Proposition 5.1 denies this, and Chapter 5 sets out why. This is not a technical refinement, it changes what the accumulation identity can be used for, and it is the reason the identity in Core Proposition 3 is written in terms of successful formation rather than expenditure.

Measurement design. They construct a composite index. Corollary 7.1 explicitly argues against composite indices for this purpose and in favour of dimensional dashboards, because aggregation destroys the information a government needs to act.

The claim of this book is therefore narrower and more specific than it may first appear. It is not that public capabilities are investments, that is established. It is that a defined class of them satisfies the full analytical test for productive capital, including the properties that investment models omit, and that treating them accordingly changes how they are accumulated, maintained, measured, and financed.

4.8 Core Proposition 2 — The Qualification Principle

The analysis developed in this chapter leads to the book's central theoretical statement. Where Core Proposition 1 asserts that qualification is possible, Core Proposition 2 specifies the conditions under which it obtains.

> **Core Proposition 2 — The Qualification Principle**
>
> Institutional capabilities qualify as Institutional Capital when they:

1. exist as accumulated stocks;
2. are created through investment;
3. constitute durable assets;
4. generate recurring institutional services;
5. require maintenance;
6. depreciate over time;
7. complement other productive assets;
8. enhance productive capacity and thereby contribute to public value.

Derived from Axioms 1 through 5.

This principle establishes the theoretical legitimacy of Institutional Capital as an analytical category.

It also provides an operational criterion for distinguishing Institutional Capital from institutional arrangements that lack these characteristics.

Two corollaries follow.

> **Corollary 2.1.** Temporary administrative activities do not constitute Institutional Capital.
>
> **Corollary 2.2.** Institutional reforms create Institutional Capital only when they generate durable, productive capabilities.

The corresponding empirical prediction is:

> **Hypothesis H2.** Institutional reforms that emphasise durable capability formation yield greater long-term administrative productivity than reforms focused solely on procedural change.

4.9 Chapter Conclusion

This chapter has demonstrated that Institutional Capital satisfies the analytical properties traditionally associated with productive capital.

The argument developed thus far has therefore established three progressively stronger conclusions.

First, certain institutional capabilities constitute durable institutional assets.

Second, these assets accumulate into Institutional Capital.

Third, Institutional Capital satisfies the defining characteristics of productive capital recognised in economic theory.

The remaining chapters no longer ask whether Institutional Capital exists.

Instead, they investigate how it functions.

The theory now shifts from qualification to economic dynamics.

Part III — Economic Dynamics